



10273 - Unveiling the nature of the most distant LyC leaker: pure star formation or AGN-boosted?

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Ion3 observations				
	1	Ion3 G235M/F170LP	NIRSpec IFU Spectroscopy	(1) Ion3
	2	Ion3 imaging	NIRCam Imaging	(1) Ion3

ABSTRACT

The quest to identify the systems responsible for the re-ionization of the Universe remains a topic of ongoing debate, with star-forming galaxies and AGNs both posited as potential drivers. While JWST has opened new avenues for exploring the high-redshift Universe, it cannot directly investigate Lyman continuum (LyC) radiation at $z > 5$. We propose spectroscopic observations using JWST/NIRSpec IFU (+ NIRCam imaging) of the most distant confirmed LyC-emitting galaxy, designated Ion3, at $z = 4.0$. Its rest-optical spectrum, observed with medium resolution G235M/F170LP ($R \sim 1000$), will:

- (i) determine whether AGN activity contributes to LyC escape in conjunction with an already observed young starburst, indicated by the presence of a broad component in the H β line at a high signal-to-noise ratio (SNR ~ 35) and/or by other line diagnostics ([NeV]3426, [OIII]4363/Hg);
- (ii) constrain the ongoing burst by its age (from EWs), the ionising photons production efficiency (from H β /UV), the star formation rate (SFR) and its surface density (the latter by leveraging NIRCam imaging);
- (iii) provide insights into the interstellar medium (ISM) properties (e.g., metallicity, ionization state) of the system, utilizing multiple lines ([OII], [NeIII], Hg, H β , [OIII], [HeI]), all observed with SNR > 3 .

Ultimately, these results will present a unique opportunity to compare the physical properties of a confirmed high-redshift LyC leaker with other low- and high-redshift leakers. Additionally, they will allow us to evaluate whether the indirect tracers of LyC escape, commonly used to characterize $z > 5$ galaxies but calibrated based on low-redshift systems, are appropriate for describing the target galaxy.

OBSERVING DESCRIPTION

We propose JWST/NIRSpec observation of the most distant star-forming galaxy with confirmed Lyman continuum leakage, at $z = 4.0$ dubbed Ion3. At the moment all we know about the physical properties of Ion3 comes from the rest-frame UV part of the spectrum. This observation aims to get

access to a plethora of rest-frame optical nebular emission lines ([OIII], H β , H γ , H δ , [NeIII], [OII]) newer before observed for LyC leaker at this distance, only 0.6 Gyr after EoR ended.

We propose two NIRSpec IFU configurations, G235M/F170LP medium resolution ($R \sim 1000$, $dv \sim 100$ km/s). The goal of the LP instrument configuration is mainly to observe the H β line and to investigate the possible presence of a broad AGN component ($dv > 500$ km/s). The "narrow" component of the H β will be used to characterize the ongoing SF event, and to compare it to archival SFR(UV), to infer the burstiness. The IFU is used to spatially characterise the morphology of the emission: two UV clumps ($R_{\text{eff}} < 200$ pc) and ~ 6 faint regions are known from previous HST photometry

Required time exposures are determined from SED, which involves fitting the ground-based VLT and space-based Spitzer photometry. Prediction for the strength of the H α line we get from the Spitzer IRAC 3.6 μ m band, where an obvious jump in flux from photometry is evident. Continuum brightness estimates are driven by SED shape in the rest-frame optical regime and it is estimated to be 23.6 mag_{AB}. Estimates from ETC indicate that we can achieve a continuum level with SNR ~ 3 and all line swith SNR > 5 .

Proposal 10273 - Targets - Unveiling the nature of the most distant LyC leaker: pure star formation or AGN-boosted?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Ion3	RA: 22 48 56.7140 (342.2363083d)		
			Dec: -44 34 46.10 (-44.57947d)		
			Equinox: J2000		
	Comments: No comment. Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies]				

Proposal 10273 - Observation 1 - Unveiling the nature of the most distant LyC leaker: pure star formation or AGN-boosted?

Observation	Proposal 10273, Observation 1: Ion3 G235M/F170LP											Fri Mar 13 19:04:24 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSspec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	Ion3	RA: 22 48 56.7140 (342.2363083d) Dec: -44 34 46.10 (-44.57947d) Equinox: J2000									
	Comments: No comment. Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		8			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235M/F170LP	NRSIRS2	10	1	false	true	NONE	8	8	5952.267	

Proposal 10273 - Observation 2 - Unveiling the nature of the most distant LyC leaker: pure star formation or AGN-boosted?

Observation	Proposal 10273, Observation 2: Ion3 imaging										Fri Mar 13 19:04:24 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	Ion3	RA: 22 48 56.7140 (342.2363083d) Dec: -44 34 46.10 (-44.57947d) Equinox: J2000 Comments: No comment. Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies]								
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	MEDIUM8	10	1	3	3	3156.61		
Special Requirements	Offset -15.743952110662175 arcsec, -18.66501767571581 arcsec Fiducial Point Override NRCBS_FULL										